

# Empirical and Molecular Formulas

Answer key with worked solutions

## A. Understanding

1. Empirical vs molecular formula.

The **empirical** formula gives the simplest whole-number ratio of atoms; the **molecular** formula gives the actual number of each kind of atom in one molecule.

2. Empirical formulas.

(a) CH (b) CH<sub>2</sub> (c) HO (d) C<sub>4</sub>H<sub>9</sub> (e) NO<sub>2</sub>

3. What is wrong with rounding 1 : 1.50?

1.50 is not close to a whole number, so it must not be rounded. Multiply both values by 2: 2 : 3, giving A<sub>2</sub>B<sub>3</sub>.

## B. Empirical formulas from percent composition

In a 100 g sample, each percentage becomes a mass in grams.

4. 85.63% C, 14.37% H.

C:  $85.63 \text{ g} \div 12.01 \text{ g/mol} = 7.130 \text{ mol}$ ; H:  $14.37 \text{ g} \div 1.008 \text{ g/mol} = 14.26 \text{ mol}$ . Divide by 7.130 mol: C = 1.00, H = 2.00. Empirical formula CH<sub>2</sub>.

5. 40.04% Ca, 12.00% C, 47.96% O.

Ca:  $40.04 \text{ g} \div 40.08 \text{ g/mol} = 0.9990 \text{ mol}$ ; C:  $12.00 \text{ g} \div 12.01 \text{ g/mol} = 0.9992 \text{ mol}$ ; O:  $47.96 \text{ g} \div 16.00 \text{ g/mol} = 2.998 \text{ mol}$ . Divide by 0.9990 mol: 1 : 1 : 3. Empirical formula CaCO<sub>3</sub>.

6. 72.36% Fe, 27.64% O.

Fe:  $72.36 \text{ g} \div 55.85 \text{ g/mol} = 1.296 \text{ mol}$ ; O:  $27.64 \text{ g} \div 16.00 \text{ g/mol} = 1.728 \text{ mol}$ . Divide by 1.296 mol: Fe = 1.000, O = 1.333. A ratio ending in .33 means multiply by 3: 3 : 4. Empirical formula Fe<sub>3</sub>O<sub>4</sub>.

## C. Molecular formulas

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7.  $\text{CH}_2\text{O}$ , molar mass 60.05 g/mol.

Empirical formula mass =  $12.01 \text{ g/mol} + 2(1.008 \text{ g/mol}) + 16.00 \text{ g/mol} = 30.03 \text{ g/mol}$ .  $n = 60.05 \text{ g/mol} \div 30.03 \text{ g/mol} = 2.00$ . Molecular formula  $\text{C}_2\text{H}_4\text{O}_2$ .

8.  $\text{CH}$ , molar mass 26.04 g/mol.

Empirical formula mass =  $12.01 \text{ g/mol} + 1.008 \text{ g/mol} = 13.02 \text{ g/mol}$ .  $n = 26.04 \text{ g/mol} \div 13.02 \text{ g/mol} = 2.00$ . Molecular formula  $\text{C}_2\text{H}_2$ .

## D. Challenge

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9. 30.43% N, molar mass 92.02 g/mol.

$\text{O} = 100\% - 30.43\% = 69.57\%$ . In 100 g: N:  $30.43 \text{ g} \div 14.01 \text{ g/mol} = 2.172 \text{ mol}$ ; O:  $69.57 \text{ g} \div 16.00 \text{ g/mol} = 4.348 \text{ mol}$ . Ratio N : O = 1 : 2.00, so the empirical formula is  $\text{NO}_2$  (46.01 g/mol).

$n = 92.02 \text{ g/mol} \div 46.01 \text{ g/mol} = 2.00$ , so the molecular formula is  $\text{N}_2\text{O}_4$ .